

Ex. No: 6

Date:

Use Sleuth Kit to analyze digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official Sleuth Kit website or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

```
arduino  
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

```
arduino  
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

3. Analyze File System with fls:

- Run the command:

```
arduino  
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

3. Recover Deleted Files with icat:

- To extract a specific file, use icat:

```
css  
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

```
css  
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

```
css  
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

```
css  
mactime -b body.txt > timeline.txt
```

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

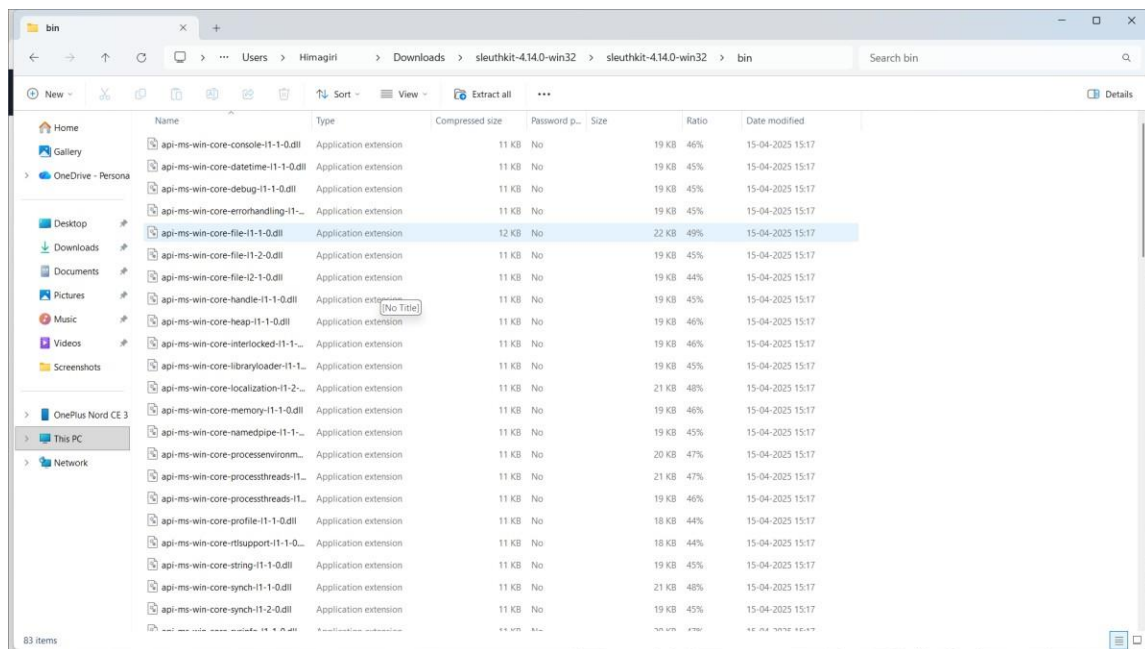
- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Fsstat command execution:



Name	Type	Compressed size	Password protected	Size	Ratio	Date modified
api-ms-win-core-console-l1-1-0.dll	Application extension	11 KB	No	19 KB	46%	15-04-2025 15:17
api-ms-win-core-datetime-l1-1-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17
api-ms-win-core-debug-l1-1-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17
api-ms-win-core-errorhandling-l1-1-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17
api-ms-win-core-file-l1-1-0.dll	Application extension	12 KB	No	22 KB	49%	15-04-2025 15:17
api-ms-win-core-file-l1-2-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17
api-ms-win-core-file-l2-1-0.dll	Application extension	11 KB	No	19 KB	44%	15-04-2025 15:17
api-ms-win-core-handle-l1-1-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17
api-ms-win-core-heap-l1-1-0.dll	Application extension	11 KB	No	19 KB	46%	15-04-2025 15:17
api-ms-win-core-interlocked-l1-1-0.dll	Application extension	11 KB	No	19 KB	46%	15-04-2025 15:17
api-ms-win-core-libraryloader-l1-1-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17
api-ms-win-core-localization-l1-2-0.dll	Application extension	11 KB	No	21 KB	48%	15-04-2025 15:17
api-ms-win-core-memory-l1-1-0.dll	Application extension	11 KB	No	19 KB	46%	15-04-2025 15:17
api-ms-win-core-namedpipe-l1-1-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17
api-ms-win-core-processenvironment-l1-1-0.dll	Application extension	11 KB	No	20 KB	47%	15-04-2025 15:17
api-ms-win-core-processthreads-l1-1-0.dll	Application extension	11 KB	No	21 KB	47%	15-04-2025 15:17
api-ms-win-core-processthreads-l1-1-0.dll	Application extension	11 KB	No	19 KB	46%	15-04-2025 15:17
api-ms-win-core-profile-l1-1-0.dll	Application extension	11 KB	No	18 KB	44%	15-04-2025 15:17
api-ms-win-core-rtlsupport-l1-1-0.dll	Application extension	11 KB	No	18 KB	44%	15-04-2025 15:17
api-ms-win-core-string-l1-1-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17
api-ms-win-core-synch-l1-1-0.dll	Application extension	11 KB	No	21 KB	48%	15-04-2025 15:17
api-ms-win-core-synch-l1-2-0.dll	Application extension	11 KB	No	19 KB	45%	15-04-2025 15:17

Identifying File System:

```
C:\Users\Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat
Missing image name
usage: fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
-t: display type only
-i imgtype: The format of the image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
-f fstype: File system type (use '-f list' for supported types)
-o imgoffset: The offset of the file system in the image (in sectors)
-P pooltype: Pool container type (use '-P list' for supported types)
-B pool_volume_block: Starting block (for pool volumes only)
-v: verbose output to stderr
-V: Print version
-k password: Decryption password for encrypted volumes
```

Installed Sleuth Kit:

```
C:\Users\Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -i ewf -o 63 -r "C:\Users\Reddy\Downloads\4Dell Latitude CP1.E0"
1*
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB1809B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 8-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident
```

Identifying File System:

```
C:\Users\Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf -o 63 -r "C:\Users\Reddy\Downloads\4Dell Latitude CP1.E0"
1* > file_lists.txt
```

```
k/d 3671-144-7: Documents and Settings
+ d/d 3673-144-7: All Users
++ d/d 3707-144-7: Application Data
+++ r/r 3800-128-1: desktop.ini
+++ d/d 3708-144-7: Microsoft
++++ d/d 3709-144-1: Crypto
+++++ d/d 3712-144-1: DSS
+++++ d/d 3713-144-1: MachineKeys
+++++ d/d 3710-144-1: RSA
+++++ d/d 3711-144-1: MachineKeys
++++ d/d 10977-144-1: Dr Watson
+++++ r/r 10979-128-3: drutsn32.log
+++++ r/r 10980-128-3: user.dmp
++++ d/d 7325-144-1: HTML Help
++++ d/d 7287-144-7: Media Index
+++++ r/r 6074-128-5: umplibary_v_0_12.db
++++ d/d 4759-144-1: Network
+++++ d/d 4762-144-1: Connections
+++++ d/d 4763-144-1: Pbk
+++++ r/r 10982-128-1: casphone.pbk
+++++ r/r 4765-128-4: sharedaccess.ini
++++ d/d 5663-144-7: User Account Pictures
+++++ d/d 5664-144-7: Default Pictures
+++++ r/r 5665-128-4: airplane.bmp
+++++ r/r 5666-128-4: astronaut.bmp
+++++ r/r 5667-128-4: ball.bmp
+++++ r/r 5674-128-4: beach.bmp
+++++ r/r 5668-128-4: butterfly.bmp
+++++ r/r 5675-128-4: car.bmp
+++++ r/r 5669-128-4: cat.bmp
+++++ r/r 5676-128-4: chess.bmp
+++++ r/r 5677-128-4: dirt bike.bmp
+++++ r/r 5678-128-4: dog.bmp
+++++ r/r 5679-128-4: drip.bmp
+++++ r/r 5680-128-4: duck.bmp
+++++ r/r 5670-128-4: fish.bmp
+++++ r/r 5681-128-4: frog.bmp
+++++ r/r 5672-128-4: guitar.bmp
+++++ r/r 5682-128-4: horses.bmp
+++++ r/r 5683-128-4: kick.bmp
+++++ r/r 5684-128-4: lift-off.bmp
+++++ r/r 5685-128-4: palm tree.bmp
+++++ r/r 5671-128-4: nink flower.hmn
```

Recovering Deleted File:

```
C:\Users\Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -i ew
f -o 63 "C:\Users\Reddy\Downloads\sleuthkit-4.14.0-win32\4Dell Latitude CPi (1).E01"
150 > recovered.txt
```

Metadata:

```
REGEDIT4

[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00000507-0000-0010-8000-00AA006D2EA4}\InprocServer32]
"ThreadingModel"="Both"

[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00000514-0000-0010-8000-00AA006D2EA4}\InprocServer32]
"ThreadingModel"="Both"

[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{0000050B-0000-0010-8000-00AA006D2EA4}\InprocServer32]
"ThreadingModel"="Both"

[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00000535-0000-0010-8000-00AA006D2EA4}\InprocServer32]
"ThreadingModel"="Both"

[HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00000541-0000-0010-8000-00AA006D2EA4}\InprocServer32]
"ThreadingModel"="Both"
```

```

e Cpi (1).E01" 150
MFT Entry Header Values:
Entry: 150      Sequence: 150
$LogFile Sequence Number: 9239131
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0 ( )
Version 16 of 16943392
Created:      1999-04-24 03:52:00.000000000 (India Standard Time)
File Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
MFT Modified: 2004-08-19 22:32:11.098512000 (India Standard Time)
Accessed:     2004-08-18 05:30:00.000000000 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: ADOFRE15.REG
Parent MFT Entry: 137      Sequence: 137
Allocated Size: 1024      Actual Size: 640
Created:      1999-04-24 03:52:00.000000000 (India Standard Time)
File Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
MFT Modified: 1999-04-24 03:52:00.000000000 (India Standard Time)
Accessed:     2004-08-18 05:30:00.000000000 (India Standard Time)

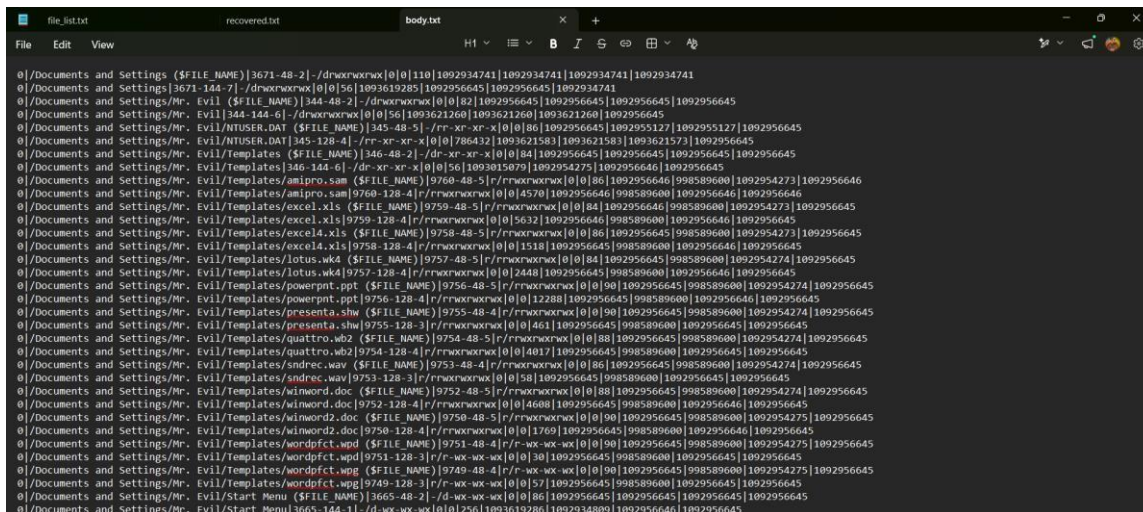
```

Generating Timeline:

```

C:\Users\Reddy\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf
-o 63 -m / -r "C:\Users\Reddy\Downloads\sleuthkit-4.14.0-win32\4Dell Latitude Cpi (1).
E01"> body.txt

```

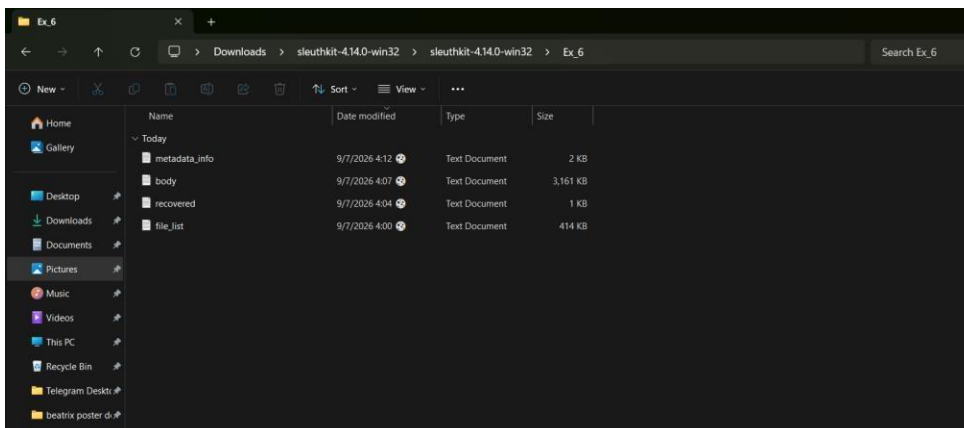


```

0:/Documents and Settings/($FILE_NAME)[3671-144-2]-/drwxrwxrwx[0][110]1092934741|1092934741|1092934741|1092934741
0:/Documents and Settings/3671-144-7[-/drwxrwxrwx[0][56]1093619285|1092956645|1092956645|1092934741
0:/Documents and Settings/Mr. Evil/($FILE_NAME)[344-48-2]-/drwxrwxrwx[0][82]1092956645|1092956645|1092956645|1092956645
0:/Documents and Settings/Mr. Evil/344-144-6[-/drwxrwxrwx[0][56]1093621260|1093621260|1093621260|1092956645
0:/Documents and Settings/Mr. Evil/NOTES.DAT ($FILE_NAME)[345-48-5]-/r-r-xr-x[0][80]1092956645|1092953127|1092953127|1092956645
0:/Documents and Settings/Mr. Evil/NOTES.DAT[345-128-4]-/r-r-xr-x[0][786432]1093621583|1093621583|1093621573|1092956645
0:/Documents and Settings/Mr. Evil/Templates ($FILE_NAME)[346-48-2]-/dr-xr-xr-x[0][84]1092956645|1092956645|1092956645|1092956645
0:/Documents and Settings/Mr. Evil/Templates[346-144-6]-/dr-xr-xr-x[0][56]1093015079|1092954275|1092956646|1092956645
0:/Documents and Settings/Mr. Evil/Templates/amipro.sam ($FILE_NAME)[9768-48-5]/r-rwxrwxrwx[0][86]1092956646|998589600|1092954273|1092956646
0:/Documents and Settings/Mr. Evil/Templates/lotus.wk4[9757-128-4]/r-rwxrwxrwx[0][2448]1092956645|998589600|1092956646|1092956645
0:/Documents and Settings/Mr. Evil/Templates/excel.xls ($FILE_NAME)[9759-48-5]/r-rwxrwxrwx[0][84]1092956646|998589600|1092954273|1092956645
0:/Documents and Settings/Mr. Evil/Templates/excel.xls[9759-128-4]/r-rwxrwxrwx[0][5632]1092956646|998589600|1092956646|1092956645
0:/Documents and Settings/Mr. Evil/Templates/excel4.xls ($FILE_NAME)[9758-48-5]/r-rwxrwxrwx[0][86]1092956645|998589600|1092954273|1092956645
0:/Documents and Settings/Mr. Evil/Templates/excel4.xls[9758-128-4]/r-rwxrwxrwx[0][1518]1092956645|998589600|1092956646|1092956645
0:/Documents and Settings/Mr. Evil/Templates/lotus.wk4[9757-128-4]/r-rwxrwxrwx[0][84]1092956645|998589600|1092954274|1092956645
0:/Documents and Settings/Mr. Evil/Templates/powerpnt.ppt ($FILE_NAME)[9756-48-5]/r-rwxrwxrwx[0][90]1092956645|998589600|1092954274|1092956645
0:/Documents and Settings/Mr. Evil/Templates/powerpnt.ppt[9756-128-4]/r-rwxrwxrwx[0][12288]1092956645|998589600|1092956646|1092956645
0:/Documents and Settings/Mr. Evil/Templates/presenta.shw ($FILE_NAME)[9755-48-4]/r-rwxrwxrwx[0][90]1092956645|998589600|1092954274|1092956645
0:/Documents and Settings/Mr. Evil/Templates/presenta.shw[9755-128-3]/r-rwxrwxrwx[0][461]1092956645|998589600|1092956645|1092956645
0:/Documents and Settings/Mr. Evil/Templates/quattro.wb2 ($FILE_NAME)[9754-48-5]/r-rwxrwxrwx[0][88]1092956645|998589600|1092954274|1092956645
0:/Documents and Settings/Mr. Evil/Templates/quattro.wb2[9754-128-4]/r-rwxrwxrwx[0][4017]1092956645|998589600|1092956645|1092956645
0:/Documents and Settings/Mr. Evil/Templates/sndrec.wav ($FILE_NAME)[9753-48-4]/r-rwxrwxrwx[0][86]1092956645|998589600|1092954274|1092956645
0:/Documents and Settings/Mr. Evil/Templates/sndrec.wav[9753-128-3]/r-rwxrwxrwx[0][58]1092956645|998589600|1092956645|1092956645
0:/Documents and Settings/Mr. Evil/Templates/winword.doc ($FILE_NAME)[9752-48-5]/r-rwxrwxrwx[0][88]1092956645|998589600|1092954274|1092956645
0:/Documents and Settings/Mr. Evil/Templates/winword.doc[9752-128-4]/r-rwxrwxrwx[0][4608]1092956645|998589600|1092956646|1092956645
0:/Documents and Settings/Mr. Evil/Templates/winword2.doc ($FILE_NAME)[9750-48-5]/r-rwxrwxrwx[0][90]1092956645|998589600|1092954275|1092956645
0:/Documents and Settings/Mr. Evil/Templates/winword2.doc[9750-128-4]/r-rwxrwxrwx[0][1769]1092956645|998589600|1092956646|1092956645
0:/Documents and Settings/Mr. Evil/Templates/wordpft.wpd ($FILE_NAME)[9751-48-4]/r-r-wx-wx-wx[0][90]1092956645|998589600|1092954275|1092956645
0:/Documents and Settings/Mr. Evil/Templates/wordpft.wpd[9751-128-3]/r-r-wx-wx-wx[0][30]1092956645|998589600|1092956645|1092956645
0:/Documents and Settings/Mr. Evil/Templates/wordpft.wpg ($FILE_NAME)[9749-48-4]/r-r-wx-wx-wx[0][90]1092956645|998589600|1092954275|1092956645
0:/Documents and Settings/Mr. Evil/Templates/wordpft.wpg[9749-128-3]/r-r-wx-wx-wx[0][857]1092956645|998589600|1092956645|1092956645
0:/Documents and Settings/Mr. Evil/Start Menu ($FILE_NAME)[3665-48-2]-/d-wx-wx-wx[0][86]1092956645|1092956645|1092956645|1092956645
0:/Documents and Settings/Mr. Evil/Start Menu[3665-144-1]-/d-wx-wx-wx[0][256]1093619286|1092934809|1092956646|1092956645

```

Gathering all the generated Files (Report):



Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The Sleuth Kit experiment successfully analyzed the provided .E01 disk image — partitions and file system structure were identified, deleted files were recovered (using fls/icat), and detailed file metadata was extracted (fsstat/istat). A timeline was generated with mactime to reconstruct file activity, and all outputs were archived with hashes to preserve evidence integrity.